

AYMANE AIT DADS

AI Research Intern | LLM Fine-Tuning · Foundation Models · NLP & Deep Learning

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PROFESSIONAL SUMMARY

Fine-tuned a 30B Mamba-Transformer MoE (Nemotron) to top 10% on a \$106K+ Kaggle competition leaderboard (0.80/1.0) via LoRA adaptation and 20+ single-variable ablation studies - directly advancing state-of-the-art reasoning with generative foundation models. Ranked 1st in class and submitted to NTIRE 2026 @ CVPR by fine-tuning CLIP ViT-L/14 (428M params) on 250K images, achieving 0.791 AUC through ensemble methods and test-time augmentation. Proficient in PyTorch, Hugging Face Transformers, and Python with hands-on experience in deep learning, natural language processing, LLMs, and statistical analysis across research-grade benchmarks. Available Summer 2026 for a 4-6 month full-time internship; currently based in France and open to relocation to Palo Alto, USA.

CORE COMPETENCIES

Foundation Models | Large Language Models | Natural Language Processing | Deep Learning | Knowledge Graphs | Hugging Face Transformers | Ablation Studies | Structured & Tabular Data

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated machine learning models (Random Forest + K-Means) on **100K+ fiber-optic network records**, classifying quality across thousands of nodes using deep feature engineering.
- Designed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Python pipelines in Scikit-learn and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, communicating research plans and results to non-technical business partners.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress, June 2026 · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE generative foundation model with LoRA ($r=32$, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types.
- Diagnosed and resolved **5 Blackwell GPU (sm_120) incompatibilities** - including monkey-patched CUDA allocator warmup, stubbed Mamba/Cutlass imports, and a pure-PyTorch RMSNorm fallback - and conducted 20+ single-variable ablations confirming completion-only loss and sequence packing as the dominant performance levers.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class

- Ranked **1st on the EURECOM private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params, pretrained on 400M image-text pairs) with a custom MLP head and dual learning-rate schedule across 250K images from 25+ generative model types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; key finding was that 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), demonstrating superior out-of-distribution generalization via a 3-model ensemble with 10-view test-time augmentation.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Statistics, Cloud Computing, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & NLP: LoRA | QLoRA | SFTTrainer | Hugging Face Transformers | PEFT | vLLM | Chain-of-Thought

ML / Deep Learning & Frameworks: PyTorch | Scikit-learn | TensorFlow | OpenCLIP | Statistical Analysis | Ensemble Methods | Ablation Studies

Engineering & Data: Python | SQL | Bash | Pandas | NumPy | Git | Linux | CUDA Debugging | Power BI